MODEL E

Muon-Decay Neutrino Communication for Submerged Platforms

Exploratory feasibility assessment | Negative practical feasibility result | 10 November 2026

Abstract. Model E investigates whether a modulated muon-decay neutrino source aboard a submerged platform could transmit low-rate information to a water Cherenkov receiver aboard a surface vessel. Simplified transmitter geometry, finite-volume detector calculations, and synthetic on-off keying (OOK) packet simulations were explored. The study did not establish a practical communication link. The illustrative source power requirement is prohibitively high, and key particle-physics and detector assumptions remain unvalidated. The numerical results are conditional sensitivity estimates, not experimentally established performance.

1. Research question and reference configuration

The illustrative scenario places a source 1,000 m below a surface receiver. The receiver is 100 metric tons of water, modeled as a cylindrical active volume of about 100 m3; a 2 m height implies a diameter of about 7.98 m. The transmitter model uses 100 MeV muon kinetic energy, a 0.2 T storage-ring field, and a 2 microsecond storage interval. A vertical-ring geometry and a selected 30-degree sector were examined. A hypothetical upward-directed decay channel was also compared. Neither a working muon source nor a working neutrino detector was designed or demonstrated.

2. Geometry and illustrative interaction model

For fixed water volume V = pi r^2 h, increasing the cylinder cross-sectional area while decreasing its height approximately preserves the number of interactions in a locally uniform neutrino flux: area times path length remains proportional to V. Finite-volume integration found only small differences among cylinder heights at kilometer-scale separations. This does not determine an optimum real detector: photon collection, containment, triggering, vessel integration and background rejection were not modeled.

The geometry scan used a constant illustrative interaction cross section of 10^-40 cm2 per free proton and perfect detection efficiency. It modeled one neutrino species per muon decay for scaling. This is **not** a validated energy-averaged cross section or a physically complete water-Cherenkov interaction model. Muon-decay flavor and energy spectra, channel thresholds, neutrino/antineutrino differences, nuclear effects, polarization and detector efficiency must be incorporated before any physical rate prediction.

Under those assumptions, at 1,000 m and a 2 m cylinder, the computed interaction probability was about 6.14 x 10^-20 per decay *conditional on the decay occurring within the selected 30-degree ring sector*. Multiplying by the sector fraction 30/360 and an approximate 0.374 probability of decay during the 2 microsecond storage period gives about 1.91 x 10^-21 illustrative interactions per stored muon. This neglects injection, transport, production and detector losses.

3. Communication model and synthetic experiments

The synthetic receiver used OOK, a four-bit sync pattern 0101, eight payload bits, and a two-bit CRC, totaling 14 transmitted bits per packet. At a 1 s bit interval, gross rate is 1 bit/s and ideal payload rate is 8/14 = 0.571 bit/s. At 0.2 s per bit, ideal payload rate is 2.857 bit/s, before losses, retransmissions or inter-packet gaps. Sampling frequencies of 10-50 Hz were explored. More samples do not themselves increase the neutrino interaction rate.

Background estimators A1 (raw mean), A2 (remove verified packet-associated peaks), A3 (also replace peaks associated with suspected but rejected packets), and four pairwise/three-way mean combinations were compared using generated events. A3 and the A2/A3 mean performed comparatively well for the deliberately strong synthetic signal; performance collapsed for weak signals. These simulations used simplified background generation, optimistic packet timing information, limited trials and heuristic peak replacement. They do **not** validate an operational BER, packet error rate or detection threshold.

4. Negative feasibility result: source power

A previous sensitivity exercise treated about 13 detected signal interactions per bit as a reference, **not a demonstrated reliability target**. At 0.2 s per bit this means 65 detected interactions/s. Dividing by the illustrative 1.91 x 10^-21 interactions per stored muon implies roughly 3.4 x 10^22 stored muons/s. At 100 MeV kinetic energy per muon, the

kinetic beam power alone is about 5.45 x 10^11 W (545 GW). This excludes muon rest energy, production inefficiency, accelerator energy, target heating, magnetic systems, shielding, cooling, and detection inefficiencies. Therefore it is a conditional lower-bound-like bookkeeping quantity, **not** a complete engineering power estimate or a reliable bound on all alternative designs.

This extreme scale is the central negative result. The proposed submarine-borne configuration cannot presently be supported as a practical engineering concept. The assumed cross section and rate calculation are insufficiently realistic to assert an exact numerical feasibility threshold, but they clearly motivate rejecting further engineering optimization until a validated end-to-end physical source-to-detection budget is available.

5. Specific failure modes and unresolved limitations

• **Unvalidated interaction physics:** fixed cross section, single-flavor proxy, no full neutrino spectra, no realistic Cherenkov thresholds or reconstruction efficiency.

• **Source realization absent:** no credible muon production rate, target yield, power conversion, injection acceptance, storage losses, or extraction-channel engineering.

• **Beam and timing assumptions:** ring-sector conditioning is not equivalent to directing all produced neutrinos; neutrinos cannot be magnetically steered after production.

• **Optimistic receiver simulation:** some tests know packet start times, use strong synthetic source rates, limited Monte Carlo trials and simplified false-alarm sampling.

• **Background model incomplete:** cosmic muons, radioactivity, optical noise, correlated bursts, vessel motion, water optics and operational trigger logic are not calibrated.

• **Shape optimization incomplete:** near-equal interaction counts across cylinder heights do not establish equivalent actual event-detection efficiencies.

• **Statistics and reproducibility:** model outputs are scenario-dependent; confidence intervals, blind searches, sensitivity tests and independent verification remain to be performed.

6. Beyond the Standard Model: speculative outlook

Supersymmetry, string-inspired extra-dimensional models and non-standard neutrino interactions could in principle modify production channels, flavor behavior or interaction cross sections. No experimentally confirmed mechanism from these ideas is known to deliver the orders-of-magnitude efficiency improvement required here. New physics may also add losses or constraints. These possibilities are included solely as speculative future research directions, not as remedies for the present feasibility failure.

7. Interpretation and status

Conclusion: negative practical feasibility for the investigated configuration. The work is an educational, reproducible exploration of a hypothesis, not a validated neutrino communication demonstration, peer-reviewed physics prediction or deployment proposal. Its useful outcome is an explicit accounting of assumptions and a strong indication that source energy, not modest refinements in receiver shape or decoding thresholds, dominates the problem. Future reassessment would require independently verified neutrino cross sections, an end-to-end source and detector model, blind acquisition simulations, and a credible system-level power budget.

8. Reproducibility and provenance

The accompanying repository includes Python scripts and CSV outputs from the exploratory simulations and geometry scans. Some scripts use fixed absolute /mnt/data output paths and require editing for local execution. Dependencies include NumPy and SciPy; no experimental data are included. Results may differ with random seeds and model choices. The work was developed iteratively in a human-AI research conversation and has not undergone independent scientific peer review.

References for independent validation: Particle Data Group Review of Particle Physics (muon and neutrino properties); Super-Kamiokande experiment publications (water Cherenkov detection); and current neutrino cross-section reviews. These are suggested background sources, not sources used to calibrate the toy cross section in this report.